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(a) Let $\gcd(a+b, a^2-ab+b^2) = k$.

Then $k \mid (a+b)$ and $k \mid (a^2-ab+b^2)$

$$\begin{aligned} \text{ie } k \mid (a+b)^2 &\Rightarrow k \mid (a^2+2ab+b^2) \\ &\Rightarrow k \mid ((a^2+2ab+b^2) - (a^2-ab+b^2)) \\ &\Rightarrow k \mid 3ab \\ &\Rightarrow \gcd(a+b, a^2-ab+b^2) \leq \gcd(a+b, 3ab) \end{aligned}$$

Let $\gcd(a+b, 3ab) = h$

Then $h \mid (a+b)$ and $h \mid 3ab$

$$\begin{aligned} \text{ie } h \mid (a+b)^2 &\Rightarrow h \mid (a^2+2ab+b^2) \\ &\Rightarrow h \mid ((a^2+2ab+b^2) - 3ab) \\ &\Rightarrow h \mid (a^2-ab+b^2) \\ &\Rightarrow \gcd(a+b, 3ab) \leq \gcd(a+b, a^2-ab+b^2) \end{aligned}$$

Hence $\gcd(a+b, a^2-ab+b^2) = \gcd(a+b, 3ab)$.

Let $s = \gcd(a+b, a^2-ab+b^2) = \gcd(a+b, 3ab)$

Then $s \mid (a+b)$ and $s \mid 3ab$

If $s \mid a$, then $s \mid ((a+b) - a) \Rightarrow s \mid b \Rightarrow s = 1$ since $\gcd(a, b) = 1$.

Similarly, if $s \mid b$, then $s \mid ((a+b) - b) \Rightarrow s \mid a \Rightarrow s = 1$ since $\gcd(a, b) = 1$.

If $s \nmid a$ and $s \nmid b$, then from $s \mid 3ab$ and $\gcd(a, b) = 1$, we have $s \mid 3$.

Therefore $s = 1$ or 3 ie $\gcd(a+b, a^2-ab+b^2) = 1$ or 3 .

(b) Let $\gcd(n!+1, (n+1)!+1) = k$

$$k \mid (n!+1) \Rightarrow n!+1 = sk \text{ for some } s \in \mathbb{Z}$$

$$k \mid ((n+1)!+1) \Rightarrow (n+1)!+1 = tk \text{ for some } t \in \mathbb{Z}$$

$$\begin{aligned} &\Rightarrow (n!)(n+1)+1 = tk \\ &\Rightarrow (sk-1)(n+1)+1 = tk \\ &\Rightarrow skn + sk - n - 1 + 1 = tk \\ &\Rightarrow n = k(sn + s - t) \\ &\Rightarrow k \mid n \\ &\Rightarrow k \mid n! \\ &\Rightarrow k \mid ((n!+1) - n!) \\ &\Rightarrow k \mid 1 \end{aligned}$$

Therefore $k = 1$ ie $\gcd(n!+1, (n+1)!+1) = 1$.

2	(i) $f(x) = \sum_{r=1}^n (p_r x - q_r)^2$ $= \sum_{r=1}^n (p_r^2 x^2 - 2p_r q_r x + q_r^2)$ $= \left(\sum_{r=1}^n p_r^2 \right) x^2 - \left(2 \sum_{r=1}^n p_r q_r \right) x + \left(\sum_{r=1}^n q_r^2 \right)$ Since $(p_r x - q_r)^2 \geq 0$, $f(x) \geq 0$ for all x. Therefore, discriminant ≤ 0. $\left(2 \sum_{r=1}^n p_r q_r \right)^2 - 4 \left(\sum_{r=1}^n p_r^2 \right) \left(\sum_{r=1}^n q_r^2 \right) \leq 0$ $\left(\sum_{r=1}^n p_r^2 \right) \left(\sum_{r=1}^n q_r^2 \right) \geq \left(\sum_{r=1}^n p_r q_r \right)^2$ Equality holds if and only if the graph of $y = f(x)$ touches the x-axis, i.e. $p_r x - q_r = 0$ for $r = 1, 2, \dots, n \Leftrightarrow x = \frac{q_1}{p_1} = \frac{q_2}{p_2} = \dots = \frac{q_n}{p_n}$. (ii) Using part (i) result with $n = 4$, $\left[\left(\frac{a}{\sqrt{cd}} \right)^2 + \left(\frac{b}{\sqrt{ad}} \right)^2 + \left(\frac{c}{\sqrt{ab}} \right)^2 + \left(\frac{d}{\sqrt{bc}} \right)^2 \right] \left[(\sqrt{cd})^2 + (\sqrt{ad})^2 + (\sqrt{ab})^2 + (\sqrt{bc})^2 \right] \geq (a+b+c+d)^2$ $\Leftrightarrow \left(\frac{d^2}{bc} + \frac{a^2}{cd} + \frac{b^2}{ad} + \frac{c^2}{ab} \right) (ab+bc+cd+da) \geq (a+b+c+d)^2$ $\Leftrightarrow (ad^3+ba^3+cb^3+dc^3)(ab+bc+cd+da) \geq abcd(a+b+c+d)^2$ (iii) By AM-GM inequality, $a+b+c+d \geq 4 \cdot (abcd)^{\frac{1}{4}}$. Therefore, $abcd(a+b+c+d)^2 \geq abcd \cdot 16(abcd)^{1/2} = 16(abcd)^{3/2}$, and the result follows. For equality to hold for the AM-GM inequality, $a = b = c = d$. For equality to hold for the Cauchy-Schwarz inequality, $\frac{cd}{a} = \frac{ad}{b} = \frac{ab}{c} = \frac{bc}{d}$, which will be satisfied when $a = b = c = d$.
3	(a) $\int \frac{\cos x}{\sin x - \cos^2 x} dx = \int \frac{\cos x}{\sin^2 x + \sin x - 1} dx$ $= \int \frac{\cos x}{\left(\sin x + \frac{1}{2} \right)^2 - \frac{5}{4}} dx$ $= \frac{1}{2 \left(\frac{\sqrt{5}}{2} \right)} \ln \left \frac{\sin x + \frac{1}{2} - \frac{\sqrt{5}}{2}}{\sin x + \frac{1}{2} + \frac{\sqrt{5}}{2}} \right + c = \frac{1}{\sqrt{5}} \ln \left \frac{2 \sin x + 1 - \sqrt{5}}{2 \sin x + 1 + \sqrt{5}} \right + c$

(b)

$$\begin{aligned}\int [f'(x)]^2 [f(x)]^N dx &= \int f'(x) \cdot f'(x) [f(x)]^N dx \\&= \frac{1}{N+1} [f(x)]^{N+1} f'(x) - \frac{1}{N+1} \int ([f(x)]^{N+1} f''(x)) dx \quad [\text{integrate by parts}] \\&= \frac{1}{N+1} [f(x)]^{N+1} f'(x) - \frac{1}{N+1} \int (\lambda f'(x) [f(x)]^{N+2}) dx \\&= \frac{1}{N+1} [f(x)]^{N+1} f'(x) - \frac{\lambda}{(N+1)(N+3)} [f(x)]^{N+3} + c\end{aligned}$$

$f(x) = \tan 4x$, $f'(x) = 4\sec^2 4x$, $f''(x) = 32\sec^2 4x \tan 4x = 8f'(x)f(x)$. So $\lambda = 8$.

$$\begin{aligned}\int_0^{\frac{\pi}{16}} \frac{\sin^{246} 4x}{\cos^{250} 4x} dx &= \frac{1}{16} \int_0^{\frac{\pi}{16}} [4\sec^2 4x]^2 (\tan 4x)^{246} dx \\&= \frac{1}{16} \left[\frac{1}{247} \tan^{247} 4x \cdot 4\sec^2 4x - \frac{8 \tan^{249} 4x}{247 \times 249} \right]_0^{\frac{\pi}{16}} \\&= \frac{1}{16} \left[\frac{1}{247} (1)(4)(2) - \frac{8}{247 \times 249} \right] \\&= \frac{1}{2} \left(\frac{249-1}{247 \times 249} \right) = \frac{124}{61503}\end{aligned}$$

4

($\Leftarrow$) If $a \mid b \Rightarrow b = ak$ for some $k \in \mathbb{Z}$.

Let $x = a$ and $y = b = ak$

Then $hcf(x, y) = (a, ak) = a$ and $lcm(x, y) = lcm(a, ak) = ak = b$.

($\Rightarrow$) There exists integers x, y such that $hcf(x, y) = a$ and $lcm(x, y) = b$.

Then $x = as$ and $y = at$ where $\gcd(s, t) = 1$

$lcm(x, y) = lcm(as, at) = b \Rightarrow b = ast$

Therefore a is a divisor of b .

Let $\frac{b}{a} = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$ where $p_1, p_2, \dots, p_r$ are the r distinct prime factors of $\frac{b}{a}$ and $k_1, k_2, \dots, k_r$ are non-negative integers. Then $b = ap_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$

$hcf(x, y) = a$ and $lcm(x, y) = b \Rightarrow x = as, y = at, b = ast$ where $\gcd(s, t) = 1$.

$$ast = ap_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$$

$$st = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$$

Since $\gcd(s, t) = 1$, then s is the product of some prime factors each of the form $p_i^{k_i}$ where t is the product of the remaining prime factors of the form $p_j^{k_j}$

Each $p_i^{k_i}$ may or may not be a factor of s which means that there are two choices for each $p_i^{k_i}$ with regard to s . Thus the total number of possible integers s is 2^r with t taking the leftovers.

Hence the number of distinct ordered pairs of positive integers (x, y) such that $hcf(x, y) = a$ and $lcm(x, y) = b$ is 2^r .

5	(i)(a) $3^{20} = 3,486,784,401$ (i)(b) $3 \times 2^{19} = 1,572,864$ (ii)(a) No. of ways to distribute the muffins to a student $= \binom{3}{1} + \binom{3}{2} + \binom{3}{3} = 7 \text{ (OR } 2^3 - 1)$ Answer = $7^6 = 117,649$ (ii)(b) No. of ways to distribute 4 identical objects (muffins) into 3 distinct boxes (flavours) $= \binom{4+2}{2} = 15 \text{ (OR, } 3 + 2 \times \binom{3}{2} + \binom{3}{2} + 3 = 15)$ Answer = $15 \times 14 \times \cdots \times 10 = 3,603,600 \text{ (OR, } {}^{15}P_6)$ (iii) Total number of ways to throw away the muffins = $\binom{6+3}{3} \binom{7+3}{3} \binom{8+3}{3} = 1,663,200$ Let A_i be the set comprising of all possible combinations the muffins can be disposed into the bins, such that auntie did not throw any muffin into the i^{th} bin. $A_i = \binom{6+2}{2} \binom{7+2}{2} \binom{8+2}{2} = 46,110$ $A_i \cup A_j = \binom{6+1}{1} \binom{7+1}{1} \binom{8+1}{1} = 504 \text{ (OR } 7 \times 8 \times 9)$ $A_i \cup A_j \cup A_k = \binom{6+0}{0} \binom{7+0}{0} \binom{8+0}{0} = 1$ Answer = $1,663,200 - \binom{4}{1} A_i + \binom{4}{2} A_i \cup A_j - \binom{4}{3} A_i \cup A_j \cup A_k$ $= 1,484,780$
6	(i)(a) $L_{n,n} = 1$ $L_{n,1} = n!$ (i)(b) First, choose 2 objects from the n distinct objects to be “together” in a box. This gives us $\binom{n}{2}$ ways of doing do. Next, we can give ordering to the 2 chosen objects within that particular box. By multiplicative principle, we have $\binom{n}{2} \times 2! = n^2 - n$ ways

(ii)(c)

Case 1: Object A is alone in a box

We can then distribute the remaining $n-1$ objects into the remaining $k-1$ boxes. This gives us $L_{n-1,k-1}$ ways of doing so.

Case 2: Object A is not alone

First, we set aside this object A and arrange the remaining $n-1$ books into the k boxes. We have $L_{n-1,k}$ ways of doing so.

Next, we want to slot this object A amongst the $n-1$ books in the k boxes. Note that we can put this object A on top (to the left) of any of the $n-1$ objects or we can also put object A right at the bottom (or rightmost) of any of the k objects. So, total of $(n+k-1)$ ways of slotting in this object A. By multiplicative principle, we have $(n+k-1)L_{n-1,k}$ for case 2.

Note that case 1 and case 2 are mutually exclusive. By addition principle, we conclude that $L_{n,k} = L_{n-1,k-1} + (n+k-1)L_{n-1,k}$

(ii)(a)

Where ordering within each box matters, we have $l_{n,k} = (k!)L_{n,k}$

$$\text{ie. } L_{n,k} = \frac{l_{n,k}}{k!}$$

From (iii),

$$L_{n,k} = L_{n-1,k-1} + (n+k-1)L_{n-1,k}$$

$$L_{n,k} = L_{n-1,k-1} + (n+k-1)L_{n-1,k}$$

$$\frac{l_{n,k}}{k!} = \frac{l_{n-1,k-1}}{(k-1)!} + (n+k-1) \frac{l_{n-1,k}}{k!}$$

Therefore, $l_{n,k} = k l_{n-1,k-1} + (n+k-1)l_{n-1,k}$, $n \geq k \geq 2$

(ii)(b) Dividing by $n!$, we remove the permutations of distinction of objects as well as the ordering of objects within each box.

Hence $\frac{l_{n,k}}{n!}$ gives the number of ways to distribute n identical objects into k distinct boxes, such that no box is empty.

To ensure no boxes are empty, we first place one object into each of the k boxes. We then distribute the remaining $(n-k)$ objects, this is akin to arrangement $(n-k)$ zeros and $(k-1)$ ones, thus giving

$$\binom{(n-k) + (k-1)}{k-1} \text{ ways.}$$

$$\text{Therefore, } \frac{l_{n,k}}{n!} = \binom{n-1}{k-1}.$$

(ii)(c) Let A be the set of all possible permutations of n distinct elements into k ordered partitions, where partitions may be empty.

For the case where $n=5$ and $k=3$, $((2,1),(3),(5,4))$ means box 1, 2 and 3 contain objects 1 and 2, object 3, objects 4 and 5 respectively. In terms of arrangements within each boxes, in box 1, object 2 comes first before object 1; whereas in box 3, object 5 comes first before object 4.

We can see that $|A| = \sum_{j=1}^k \binom{k}{j} l_{n,j} = n! \sum_{j=1}^k \binom{k}{j} \binom{n-1}{j-1}$ which counts the number of ways to distribute n distinct objects into k distinct boxes such that the ordering of objects within each box is important and that boxes may be empty.

Let B be all possible permutations of the $(n+k-1)$ numbers involving n distinct integers from $\{1, 2, \dots, n\}$, as well as $(k-1)$ zeros. We can see that there exists a bijection f between A and B . In the case where $n=5$ and $k=3$,

$$f((2,1), (3), (5,4)) = 2103054$$

$$f((1,3,5,4), (), (2)) = 1354002$$

Now, arranging $\{1, 2, \dots, n; \underbrace{0, 0, \dots, 0}_k\}$ gives us $|B| = \frac{(n+k-1)!}{(k-1)!}$.

Since $|A| = |B|$, we have $n! \sum_{j=1}^k \binom{k}{j} \binom{n-1}{j-1} = \frac{(n+k-1)!}{(k-1)!}$.

Therefore, $\sum_{j=1}^k \binom{k}{j} \binom{n-1}{j-1} = \frac{(n+k-1)!}{n!(k-1)!}$.

7

(i) Let P_n be the statement $2u_n^2 - 3v_n^2 + 6u_nv_n = 5$ for $n \in \mathbb{Z}^+$.

For $n=1$, $\text{LHS} = 2u_1^2 - 3u_1^2 + 6u_1v_1 = 2(1) - 3(1) + 6 = 5 = \text{RHS}$. So P_1 is true.

Assume that P_k is true for some $k \in \mathbb{Z}^+$, i.e. $2u_k^2 - 3v_k^2 + 6u_kv_k = 5$.

$$\begin{aligned} \text{LHS of } P_{k+1} &= 2u_{k+1}^2 - 3v_{k+1}^2 + 6u_{k+1}v_{k+1} \\ &= 2(u_k^2 + 6u_kv_k + 9v_k^2) - 3(4u_k^2 + 28u_kv_k + 49v_k^2) + 6(2u_k^2 + 13u_kv_k + 21v_k^2) \\ &= 2u_k^2 - 3v_k^2 + 6u_kv_k \\ &= 5. \quad \text{So } P_k \text{ is true} \Rightarrow P_{k+1} \text{ is true.} \end{aligned}$$

Since P_1 is true and P_k is true $\Rightarrow P_{k+1}$ is true, by the Principle of Mathematical Induction, P_n is true for all $n \in \mathbb{Z}^+$.

[(ii) For $2u_n^2 - 3v_n^2 + 6u_nv_n = 5$, dividing throughout by v_n^2 yields $2r_n^2 - 3 + 6r_n = \frac{5}{v_n^2}$.

Since $\lim_{n \rightarrow \infty} (v_n) = \infty$ and $\lim_{n \rightarrow \infty} (r_n) = L$,

$$2L^2 - 3 + 6L = 0 \Rightarrow L = \frac{1}{2}(-3 + \sqrt{15}) \text{ or } \frac{1}{2}(-3 - \sqrt{15}).$$

However, as $r_n > 0$, its limit L must be non-negative. Therefore, $L = \frac{1}{2}(-3 + \sqrt{15})$.

$$\begin{aligned}
\text{(iii)} \quad r_{n+1} - r_n &= \frac{u_n + 3v_n}{2u_n + 7v_n} - \frac{u_n}{v_n} \\
&= \frac{u_n v_n + 3v_n^2 - 2u_n^2 - 7u_n v_n}{v_n(2u_n + 7v_n)} \\
&= -\frac{2u_n + 6u_n v_n - 3v_n^2}{v_n(2u_n + 7v_n)} \\
&= -\frac{5}{v_n(2u_n + 7v_n)} < 0 \quad (\text{since } u_n, v_n > 0).
\end{aligned}$$

So the sequence $\{r_n\}$ is strictly decreasing and tends to the limit $\frac{1}{2}(-3 + \sqrt{15})$.

$$\therefore r_n > \frac{1}{2}(-3 + \sqrt{15})$$

$$\Rightarrow 2r_n + 3 > \sqrt{15} \quad \text{---- (2)}$$

$$\Rightarrow \frac{1}{2r_n + 3} < \frac{1}{\sqrt{15}}$$

$$\Rightarrow \frac{15}{2r_n + 3} < \sqrt{15} \quad \text{---- (3)}$$

By taking intersection of (2) and (3), we have the result.

$$\text{(iv)} \quad r_3 = \frac{u_3}{v_3} = \frac{31}{71}. \text{ So } \frac{15}{2r_3 + 3} < \sqrt{15} < 2r_3 + 3 \Rightarrow \frac{213}{55} < \sqrt{15} < \frac{275}{71}.$$

8 (i) $\phi(p) = p - 1$ since all the numbers $1, 2, 3, \dots, p - 1$ are all relatively prime to the prime number p .

(ii) The only positive integers not exceeding p^r which are not coprime to p^r are $1p, 2p, 3p, \dots, p^{r-1}, p^r$. As there are p^{r-1} of such integers, $\phi(p^r) = p^r - p^{r-1} = p^{r-1}(p - 1)$.

(iii) Case (1): x is a power of 2 only ie $x = 2^r, r \geq 2$.

Then $\phi(x) = \phi(2^r) = 2^{r-1}(2 - 1) = 2^{r-1}$ which is even.

Case (2) x is not a power of 2 only, then it has an odd prime p as a factor. Thus $x = p^r q$ where $r \geq 1$ and $\gcd(p^r, q) = 1$.

By the above result,

$$\phi(x) = \phi(p^r q) = \phi(p^r) \phi(q) = p^{r-1}(p - 1) \phi(q)$$

Since p is an odd prime, then $2 \mid (p - 1)$ ie $\phi(x)$ is even

Hence $\phi(x)$ is even for all $x \geq 3$.

(iv) Consider $n = 2^t 3^s m$ where t and s are non-negative integers and m is a positive integer which does not have 2 or 3 as factors, ie $\gcd(6, m) = 1$

$$\phi(3x) = \phi(2x)$$

$$\phi(3 \cdot 2^t 3^s m) = \phi(2 \cdot 2^t 3^s m)$$

$$\phi(2^t 3^{s+1} m) = \phi(2^{t+1} 3^s m)$$

$$\phi(2^t 3^{s+1}) \phi(m) = \phi(2^{t+1} 3^s) \phi(m) \text{ since } m \text{ is coprime to both } 2^t 3^{s+1} \text{ and } 2^{t+1} 3^s$$

$$\phi(2^t 3^{s+1}) = \phi(2^{t+1} 3^s)$$

$$\phi(2^t) \phi(3^{s+1}) = \phi(2^{t+1}) \phi(3^s) \text{ since powers of 2's and powers of 3's are coprime}$$

Case 1: For $s \geq 1, t \geq 1$,

$$\phi(2^t) \phi(3^{s+1}) = \phi(2^{t+1}) \phi(3^s)$$

$$2^{t-1} (2-1) 3^s (3-1) = 2^t (2-1) 3^{s-1} (3-1)$$

$$2^{t-1} 3^s = 2^t 3^{s-1}$$

$$3 = 2 \text{ (no solution)}$$

Case 2: For $s \geq 1, t = 0$,

$$\phi(2^t) \phi(3^{s+1}) = \phi(2^{t+1}) \phi(3^s)$$

$$\phi(1) \phi(3^{s+1}) = \phi(2) \phi(3^s)$$

$$3^s (3-1) = 3^{s-1} (3-1)$$

$$3 = 1 \text{ (no solution)}$$

Case 3: For $s = 0, t \geq 1$,

$$\phi(2^t) \phi(3^{s+1}) = \phi(2^{t+1}) \phi(3^s)$$

$$\phi(2^t) \phi(3) = \phi(2^{t+1}) \phi(1)$$

$$2^{t-1} (2-1) (2) = 2^t (2-1) (1)$$

$$2^t = 2^t$$

ie true for every $t \geq 1$ and $s = 0$ and positive integers m where $\gcd(6, m) = 1$.

Case 4: For $s = 0, t = 0$,

$$\phi(2^t) \phi(3^{s+1}) = \phi(2^{t+1}) \phi(3^s)$$

$$\phi(1) \phi(3) = \phi(2) \phi(1)$$

$$(1)(2) = (1)(1)$$

$$2 = 1 \text{ (no solution)}$$

Therefore the positive values of x are all even numbers which do not have 3 as a factor.